ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 1 style (theory). Section A: objective questions (30) · Section B: history, laws and classes of robots, safety and ethics, robot anatomy, degrees of freedom and grippers, open- and closed-loop control with encoders and PID, sensors, actuators and robot programming methods (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places. Use g = 9,8 m/s².
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The word "robot" was first used in a play written by …

A Isaac Asimov. B Karel Čapek.
C George Devol. D Joseph Engelberger.

1.1.2 A robot arm whose three main axes all move in straight lines (X, Y and Z) is a/an …

A Cartesian robot. B articulated robot.
C SCARA robot. D spherical robot.

1.1.3 The total space that a robot's end effector can reach is called its …

A payload. B accuracy. C work envelope. D repeatability.

1.1.4 The tool on the end of a robot arm that does the actual work is the …

A controller. B teach pendant. C end effector. D base.

1.1.5 A control system that measures its output and uses the result to correct errors is a/an …

A open-loop system. B manual system.
C closed-loop system. D timer-only system.

1.1.6 An encoder on a motor shaft measures …

A rotation and position. B temperature.
C light level. D humidity.

1.1.7 An ultrasonic sensor measures distance by timing …

A infrared light pulses. B the echo of a sound pulse.
C a magnetic field. D air pressure.

1.1.8 The ability of a robot to return to the same point again and again is its …

A speed. B payload. C reach. D repeatability.

1.1.9 Teaching a robot by physically guiding its arm through the movements is called …

A offline programming. B lead-through programming.
C text programming. D cloud programming.

1.1.10 A collaborative robot (cobot) is designed to …

A work behind a safety fence only. B work only in space.
C replace every worker in a factory. D work safely alongside people.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 According to Asimov's laws, a robot may injure a human being if it is ordered to.

1.2.2 A SCARA robot is well suited to fast pick-and-place assembly work.

1.2.3 Degrees of freedom are the number of independent movements a robot can make.

1.2.4 Payload is the maximum mass that a robot can carry at its wrist.

1.2.5 An open-loop system checks its output with sensors.

1.2.6 A stepper motor turns in fixed angular steps.

1.2.7 Vacuum grippers work best on porous objects such as sponges.

1.2.8 A teach pendant is a handheld unit used to move and program a robot.

1.2.9 Offline programming means the robot must stop production while the program is written.

1.2.10 Robots are well suited to dull, dirty and dangerous jobs.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Joint that turns about an axis	A	emergency stop
1.3.2	Joint that slides in a straight line	B	accuracy
1.3.3	Robot with a waist, shoulder and elbow, like a human arm	C	mobile robot
1.3.4	How close a robot gets to the point it was commanded to reach	D	revolute joint
1.3.5	End effector that holds objects by suction	E	vacuum gripper
1.3.6	Measured output sent back to be compared with the set point	F	offline programming
1.3.7	Sensor that measures tilt and acceleration	G	prismatic joint
1.3.8	Writing and simulating a robot program on a computer	H	feedback
1.3.9	Robot that moves around on wheels, tracks or legs	I	articulated robot
1.3.10	Large red button that stops the robot immediately	J	accelerometer

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ROBOT CLASSES, SAFETY AND ETHICS

2.1 State Asimov's THREE laws of robotics. (3)

2.2 Name THREE classes of robots according to where they are used, and give ONE example of each. (6)

2.3 Name FOUR safety measures that protect workers in an industrial robot work cell. (4)

2.4 Give ONE ethical concern about using robots in South African workplaces. (1)

[14]

QUESTION 3: ROBOT ANATOMY AND END EFFECTORS

3.1 Name and briefly describe FOUR main parts of an industrial robot system. (4)

3.2 A robot arm has a waist that rotates, a shoulder that rotates, an elbow that rotates and a wrist that can pitch, yaw and roll. How many degrees of freedom does the arm have? Explain your answer. (2)

3.3 Explain the difference between the accuracy and the repeatability of a robot. Use an example. (4)

3.4 Name TWO types of mechanical or non-mechanical grippers and give ONE object that each is best suited to pick up. (4)

3.5 A two-finger friction gripper must lift a 2 kg metal block. The coefficient of friction between the fingers and the block is 0,4. Calculate the gripping force that each finger must apply, using a safety factor of 2. (4)

[18]

QUESTION 4: CONTROL SYSTEMS AND ENCODERS

4.1 Differentiate between open-loop and closed-loop control, and give ONE robot-related example of each. (4)

4.2 Draw a neat, labelled block diagram of a closed-loop system that controls the angle of one robot joint. (5)

4.3 A mobile robot has an encoder that gives 600 pulses per revolution, fitted directly to a wheel with a diameter of 65 mm. Calculate:

4.3.1 The angle the wheel turns for each pulse (1)

4.3.2 The circumference of the wheel (2)

4.3.3 The distance the robot moves for each pulse (2)

4.3.4 The number of pulses counted when the robot drives 1 m (2)

4.4 State the purpose of the proportional (P) term and the derivative (D) term in a PID controller. (2)

[18]

QUESTION 5: SENSORS, ACTUATORS AND PROGRAMMING

5.1 An ultrasonic sensor receives its echo 2,9 ms after sending the pulse. The speed of sound is 343 m/s. Calculate the distance to the obstacle. (3)

5.2 Compare a DC motor, a stepper motor and a servo motor. For EACH, give ONE feature and ONE suitable robot use. (6)

5.3 Name the THREE main methods of programming industrial robots and give ONE advantage of EACH. (6)

5.4 Explain the difference between internal (proprioceptive) and external (exteroceptive) sensors. Give ONE example of each. (4)

5.5 Name the sensor that builds a map of the surroundings by timing reflected laser pulses. (1)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER